

Which verification directive is used to check a design property?

- ☐ assume
- ☒ cover
- ☐ assert
- ☐ restrict

The verification directive assert requires that the design behavior should be checked to be always true.

Incorrect



Which of the following is NOT a type of code coverage?

- ☐ Branch coverage
- ☒ Block coverage
- ☐ Expression coverage
- ☐ Transaction coverage

Transaction coverage is a type of functional coverage, not code coverage.

Incorrect



You have correctly answered 21 of 100 questions

Question 3 of 100

Which of the following are goals of coverage? Select all that apply.

- ☐ Predict test development completion
- ☒ Optimize design performance
- ☐ Direct further test development
- ☐ Confirm existing test efficacy

The primary goals of coverage are to confirm the efficacy of existing tests, direct further test development, and predict test development completion.

Incorrect



You have correctly answered 21 of 100 questions

Question 4 of 100

Which Cadence® product is used for processor-based emulation?

- ☐ Protium™
- ☒ Xcelium™
- ☐ Virtuoso®
- ☐ Palladium®

Palladium is Cadence's product for processor-based emulation.

Incorrect



How can you access a class member in SystemVerilog?

- ☐ Using brackets {}
- ☒ Using the # operator
- ☐ Using the @ operator
- ☐ Using the . operator

Class members are accessed using the dot (.) operator.

Incorrect



What are the benefits of formal verification over simulation? Select all that apply.

- ☐ Requires less memory
- ☒ Finds all possible bugs
- ☐ No testbench required
- ☐ Exhaustive verification

Formal verification does not require a testbench, is exhaustive, and can find all possible bugs.

Incorrect



For basic coverage of RTL design code, which coverage type would you choose and how would you collect it?

- ☐ FSM coverage, asynchronously
- ☒ Transaction coverage, synchronously
- ☐ Code coverage, asynchronously
- ☐ Code coverage, synchronously

Basic coverage of RTL design code is best achieved with code coverage collected asynchronously.

Incorrect



What is the benefit of using assertions in verification?

- ☐ They can detect bugs earlier in the design cycle
- ☒ They reduce the need for testbenches
- ☐ They eliminate the need for simulation
- ☐ They simplify the design process

Assertions can detect bugs earlier in the design cycle, improving overall verification efficiency.

Incorrect



Which of the following are verification acceleration platforms? Select all that apply.

- ☐ Single-Core Simulation
- ☒ FPGA-based Prototyping
- ☐ Multi-Core Simulation
- ☐ Processor-based Emulation

The verification acceleration platforms include Multi-Core Simulation, Processor-based Emulation, and FPGA-based Prototyping.

Incorrect



What is the main function of a UVM UVC monitor?

- ☐ To collect information from the DUT
- ☒ To generate stimulus
- ☐ To apply stimulus
- ☐ To check the outputs of the DUT

The monitor collects information directly from the DUT for checking and coverage purposes.

Incorrect



You have correctly answered 21 of 100 questions

Question 11 of 100

What is the purpose of the inside construct in SystemVerilog?

- ☐ To specify a time literal
- ☒ To perform a comparison between an expression and a list of values
- ☐ To declare a variable
- ☐ To define a new data type

The inside construct in SystemVerilog performs a comparison between an expression and a list of values.

Correct

You have correctly answered 21 of 100 questions

Question 12 of 100

What does MDV stand for?

- ☐ Method Driven Verification
- ☒ Model Driven Verification
- ☐ Metric Driven Verification
- ☐ Manual Directed Verification

MDV stands for Metric Driven Verification, which is a methodology that uses metrics to drive the verification process, ensuring that all features and scenarios are adequately tested.

Incorrect

Which of the following are new procedural blocks introduced in SystemVerilog? Select all that apply.

- ☐ always_ff
- ☒ always_comb
- ☐ always_latch
- ☐ always

SystemVerilog introduces new procedural blocks such as always_ff, always_latch, and always_comb.

Incorrect



What is the primary benefit of using random data generation in verification?

- ☐ Ensures repeatability of tests
- ☒ Simplifies code writing
- ☐ Reduces simulation time
- ☐ Generates large amounts of stimulus with minimal code

Random data generation allows for extensive testing with minimal code, enhancing the thoroughness of verification.

Incorrect



Which of the following is true about the extern keyword in SystemVerilog?

- ☐ It is used to declare external variables.
- ☒ It is used to declare external randomization constraints.
- ☐ It is used to declare external properties.
- ☐ It is used to declare external methods outside the class declaration.

The extern keyword allows the implementation of class methods to be declared outside the class declaration for readability and convenience.

Incorrect

Given the following code snippet, what will be the value of data after randomization?

```
logic [7:0] data;  
int ok;  
initial begin  
    ok = randomize(data) with ( data >= 32; data <= 126; );  
end
```

- ☐ Any value between 0 and 31
- ☒ Any value between 0 and 255
- ☐ Any value between 127 and 255
- ☐ Any value between 32 and 126

The constraint `data >= 32; data <= 126;` restricts the random values of data to be between 32 and 126.

Incorrect

The `disable iff` construct is used to terminate an assertion when a condition is true.

- ☐ True
- ☒ False

The `disable iff` construct is used to terminate an assertion when a specified condition is true.

Incorrect



Which of the following are benefits of using assertions? Select all that apply.

- ☐ Reduces the need for testbenches
- ☒ Improves observability
- ☐ Can be applied non-intrusively to legacy designs
- ☐ Can concisely document design intent

Assertions improve observability, can be applied non-intrusively to legacy designs, and can concisely document design intent.

Incorrect



Who typically writes assertions and for what purpose? Select all that apply.

- ☐ RTL Designers - Functional properties to ensure intended behavior
- ☒ Verification Engineers - Functional coverage to ensure test stimulus exercises corner cases
- ☐ System Architects - Abstract properties to ensure implementation meets functional spec
- ☐ IP Designers - Correct use of IP to indicate misuse

Different roles write assertions for various purposes, including ensuring functional specs, intended behavior, exercising corner cases, and indicating misuse.

Incorrect



What is the purpose of coverage measurement in the verification process?

- ☐ To confirm what has and has not been tested
- ☒ To define the verification plan
- ☐ To generate random stimulus
- ☐ To check for correct behavior

Coverage measurement confirms what has and has not been tested, providing feedback on the effectiveness of the verification process.

Incorrect



Which of the following statements about 2 and 4 state types in SystemVerilog are true? Select all that apply.

- ☐ You can freely assign 2 state types to 4 state types
- ☒ You cannot assign 2 state types to 4 state types
- ☐ You can freely assign 4 state types to 2 state types
- ☐ You cannot assign 4 state types to 2 state types

SystemVerilog allows direct assignment of 2 state types to 4 state types, and vice-versa. Although it is advisable to avoid mixing 2 and 4 state types.

Incorrect



Which of the following operators is used for next cycle implication in SystemVerilog assertions?

- ☐ ##1
- ☒ |=>
- ☐ |->
- ☐ &&

The |=> operator is used for next cycle implication in SystemVerilog assertions.

Correct



You have correctly answered 21 of 100 questions

Question 23 of 100

The scale of the functional verification problem is manageable with current tools.

- ☐ True
- ☒ False

The scale of the functional verification problem is massive due to the complexity of modern IC designs, making it impossible to test everything with current tools. Verification focuses on risk management rather than complete coverage.

Correct

You have correctly answered 21 of 100 questions

Question 24 of 100

What is the primary purpose of coverage in verification?

- ☐ To reduce simulation time
- ☒ To optimize the design code
- ☐ To measure the performance of the design
- ☐ To measure how well the testbench verifies the design

Coverage is used to measure the effectiveness of the testbench in verifying the design and identifying areas that need more focus.

Incorrect

The stimulus for a sequence is defined in which task method?

- ☐ main()
- ☒ run_phase()
- ☐ body()
- ☐ new()

The stimulus for a sequence is defined in a task named body().

Incorrect



Which technology uses Field Programmable Gate Arrays (FPGAs) for prototyping?

- ☐ Multi-Core Simulation
- ☒ Emulation
- ☐ Single-Core Simulation
- ☐ Rapid-Prototyping

Rapid-Prototyping uses FPGA for prototyping.

Incorrect



Which method is used to generate random values on scope variables in SystemVerilog?

- ☐ random()
- ☒ generate()
- ☐ rand()
- ☐ randomize()

The randomize() method is used to generate random values on scope variables in SystemVerilog.

Incorrect

Coverage is not important in formal analysis.

- ☐ True
- ☒ False

Coverage is as important in formal analysis as it is in simulation to ensure all states and behaviors are verified.

Correct

Which types of coverage are included in functional coverage? Select all that apply.

- ☐ Control-oriented coverage
- ☒ Block coverage
- ☐ Data-oriented coverage
- ☐ Toggle coverage

Functional coverage includes data-oriented and control-oriented coverage.

Incorrect



What is the main benefit of using UVM?

- ☐ It provides a scalable verification methodology
- ☒ It locks users into a single vendor solution
- ☐ It is a proprietary solution
- ☐ It is only useful for block-level verification

UVM provides a scalable verification methodology that can be used from block-level to system-level verification.

Incorrect



What does the following assertion check?

```
property SEQ_COUNT;  
  @(negedge clk) !a ##1 !a ##1 !a ##1 !a => a;  
endproperty  
ASC: assert property (SEQ_COUNT);
```

- ☐ a is high for four cycles and then low
- ☐ a is low for four cycles and then high
- ☐ a is high for three cycles and then low
- ☒ a is low for three cycles and then high

The assertion checks that a is low for four cycles and then high.

Incorrect



The always_comb procedural block in SystemVerilog can contain embedded timing or event control.

- ☐ True
- ☒ False

The always_comb procedural block cannot contain embedded timing or event control.

Correct



What is the primary purpose of an assertion in verification?

- ☐ To generate test vectors
- ☐ To verify that a design property is always true
- ☐ To reduce simulation time
- ☐ To optimize design performance

An assertion is a directive to an EDA tool to verify that a property is always true.

Correct

What is the main purpose of a UVM Verification Component (UVC)?

- ☐ To provide checks and coverage
- ☒ To generate stimulus for the DUT
- ☐ To be reusable and configurable
- ☐ All of the above

A UVC is designed to generate stimulus, provide checks and coverage, and be reusable and configurable.

Incorrect

Human error can lead to incomplete verification even if all checks pass.

- ☐ True
- ☒ False

Human error can result in incomplete verification because the verification plan might not cover all required functionality or might be implemented incorrectly, leading to undetected bugs.

Incorrect



What is the primary goal of verification in the context of digital design?

- ☐ To stimulate the DUT and check the DUT response
- ☒ To reduce the power consumption of the design
- ☐ To increase the performance of the design
- ☐ To minimize the area of the design

The fundamental verification activity is to stimulate the Design Under Test (DUT) and check the DUT response to ensure it behaves as expected.

Incorrect



You have correctly answered 21 of 100 questions

Question 37 of 100

Which of the following is a type of UVC specific to a protocol or bus?

- ☐ Register Model UVC
- ☒ Module UVC
- ☐ Scoreboard UVC
- ☐ Interface UVC

Interface UVCs are specific to a protocol, bus, or set of signals on the DUT.

Incorrect



You have correctly answered 21 of 100 questions

Question 38 of 100

What is the main benefit of faster simulation?

- ☐ Higher latency
- ☒ Less complete regression testing
- ☐ Quicker test-debug-fix-verify cycles
- ☐ Shorter random tests

Faster simulation reduces latency, allowing quicker test-debug-fix-verify cycles and more complete regression testing.

Incorrect



You have correctly answered 21 of 100 questions

Question 39 of 100

Which verification methodology is described as the most successful?

- ☐ Formal Verification
- ☒ Random Testing
- ☐ Directed Tests
- ☐ Metric Driven Verification (MDV)

MDV is considered the most successful verification methodology because it uses a hierarchical plan to track coverage and checks for each feature of the design, ensuring thorough verification based on experience and specifications.

Incorrect



You have correctly answered 21 of 100 questions

Question 40 of 100

Pure random number sequences are useful for verification.

- ☐ True
- ☒ False

Pure random number sequences are not useful for verification because they do not allow for repeatability, which is essential for debugging.

Correct



What does the following assertion check?

```
property ENABLE_CHK;  
@(negedge clk) en1 && en2;  
endproperty  
AEC : assert property (ENABLE_CHK);
```

- ☐ en1 and en2 are both high at the negative edge of the clock
- ☒ en1 and en2 are both low at the negative edge of the clock
- ☐ en1 and en2 are both low at the positive edge of the clock
- ☐ en1 and en2 are both high at the positive edge of the clock

The assertion checks that en1 and en2 are both high at the negative edge of the clock.

Incorrect



What is the primary benefit of using Metric-Driven Verification (MDV)?

- ☐ Improved predictability, productivity, and quality of verification
- ☒ Larger area
- ☐ Increased power consumption
- ☐ Reduced performance

MDV improves the predictability, productivity, and quality of the verification effort by using metrics and automation.

Incorrect



What are the benefits of using constrained random stimulus in verification? Select all that apply.

- ☐ Less manual effort required to determine and follow paths
- ☒ Manual effort required to check for expected results
- ☐ Manual effort required to verify that goals are reached
- ☐ Non-goal scenarios are more likely to be exercised

Constrained random stimulus reduces manual effort to determine and follow paths and increases the likelihood of exercising non-goal scenarios.

Incorrect



Which data type in SystemVerilog is a direct replacement for the Verilog reg type?

- ☐ logic
- ☒ shortint
- ☐ bit
- ☐ longint

The logic data type in SystemVerilog is a direct replacement for the Verilog reg type.

Incorrect



What is the purpose of assertions in the verification process?

- ☐ To measure coverage
- ☒ To define the verification plan
- ☐ To generate random stimulus
- ☐ To check for correct and incorrect behavior

Assertions are used to check for correct and incorrect behavior during the course of simulation, ensuring that the design reacts appropriately to the stimulus.

Incorrect



Given the following code snippet, what will happen if wr_en and rd_en are both high at the negative edge of the clock?

```
RW_CHK : assert property ( @(negedge clock) !(wr_en && rd_en) );
```

- ☐ The assertion will fail
- ☐ The assertion will be disabled
- ☐ The assertion will be ignored
- ☒ The assertion will pass

The assertion checks that wr_en and rd_en are never both high at the negative edge of the clock. If they are both high, the assertion will fail.

Incorrect



What is the primary benefit of using assertions in the verification process?

- ☐ To measure coverage
- ☒ To check for correct and incorrect behavior
- ☐ To define the verification plan
- ☐ To generate random stimulus

Assertions check for correct and incorrect behavior during simulation, ensuring the design reacts appropriately to the stimulus.

Correct



Protium™ uses the smallest FPGAs available.

- ☐ True
- ☒ False

Protium uses some of the largest FPGAs available, such as the Xilinx Vu19p FPGA.

Correct



What is the purpose of a UVM scoreboard?

- ☐ To apply stimulus
- ☒ To monitor the DUT
- ☐ To check the operation of the DUT
- ☐ To generate stimulus

The scoreboard checks the operation of the DUT by collecting input and output data and verifying their compatibility.

Incorrect



Coverage measurement helps in determining what has and has not been tested in the verification process.

- ☐ True
- ☒ False

Coverage measurement provides feedback on what has and has not been tested, making the verification process more effective.

Incorrect



What is the purpose of the new function in a SystemVerilog class?

- ☐ To define class methods
- ☒ To define class instances
- ☐ To declare class properties
- ☐ To create and initialize class instances

The new function is the constructor method used to create and initialize instances of a class.

Incorrect



If it takes a week of single core simulation to boot Linux on a design, how long would we expect this to take with emulation?

- ☐ Under a day
- ☒ A month
- ☐ An hour
- ☐ A week

You could expect simulation which takes a week on a normal single core simulation, to take under a day for emulation, depending on the design etc.

Incorrect



Which component of MDV involves analyzing checker-reported errors and providing more direction when goals are not being met?

- ☐ Planning verification goals
- ☒ Assertion definition
- ☐ Coverage measurement
- ☐ Generation of random stimulus

Planning verification goals involves analyzing checker-reported errors and providing more direction when goals are not being met.

Incorrect

A SystemVerilog logic type can be assigned from which of the following? Select all that apply.

- ☐ A single output port only
- ☒ A single always_comb or always_ff block only
- ☐ Any number of assign statements
- ☐ Any number of initial or always blocks
- ☐ Any number of always_comb or always_ff blocks
- ☐ Any combination of assigns, output ports and procedural blocks
- ☐ A single assign statement only

Any number of always or initial blocks is allowed for backward compatibility reasons, but otherwise a logic can only be assigned from a single source - assign, output port, always_ff or always_comb.

Incorrect

What is the role of the Universal Verification Methodology (UVM) in the verification process?

- ☐ To increase the performance of the design
- ☒ To optimize the design for power consumption
- ☐ To provide a class library and methodology for verification
- ☐ To reduce the area of the design

UVM provides a class library of verification building blocks and a methodology for using these blocks to create scalable and reusable verification components.

Incorrect

What does the `assume` directive do in formal verification?

- ☐ It checks that a property is always true
- ☒ It specifies a constraint on the inputs
- ☐ It ignores the property
- ☐ It counts how many times an expression is true

The 'assume' directive specifies a constraint on the inputs, informing the tool how the design inputs can behave.

Correct

Given the following code, what is the value of state at the end of the initial block?

```
typedef enum {ONE, TWO, THREE} state_t;  
state_t state;  
  
initial begin  
    state = THREE;  
    state = state.next();  
end
```

- ☐ ONE
- ☐ 3
- ☐ 4
- ☒ FOUR
- ☐ THREE

The enumerate next() method iterates over the declared values only. Therefore, when state is the last declared value THREE, state.next() wraps round to the first value ONE.

Incorrect



What does FSM coverage focus on?

- ☐ Control signal sequences
- ☒ Data signal values
- ☐ State machine states and transitions
- ☐ Transaction payloads

FSM coverage measures the states visited and transitions between states in a finite state machine.

Incorrect



Formal verification can prove properties under all input conditions allowed by constraints.

- ☐ True
- ☒ False

Formal verification proves properties under all input conditions allowed by the constraints.

Incorrect



Which of the following statements about class aggregation in SystemVerilog is true?

- ☐ Aggregation allows a class to contain instances of other classes as properties.
- ☒ Aggregation allows a class to override methods of another class.
- ☐ Aggregation allows a class to inherit properties from another class.
- ☐ Aggregation allows a class to be declared outside a module.

Aggregation refers to a class containing instances of other classes as its properties, creating a composite class.

Incorrect



What is the main challenge in scaling the verification effort for modern designs?

- ☐ Minimizing the area of the design
- ☒ Reducing the power consumption
- ☐ Keeping pace with the increasing size and complexity of the DUT
- ☐ Increasing the number of transistors

The main challenge is how to scale the verification effort to keep pace with the continually increasing size and complexity of the DUT.

Incorrect



What is the difference between a static and non-static method in SystemVerilog?

- ☒ A static method can access both static and non-static members, while a non-static method can access only non-static members.
- ☐ A static method can be called only from within the class, while a non-static method can be called from outside the class.
- ☐ A static method can access only static members, while a non-static method can access both static and non-static members.
- ☐ A static method can be called only from outside the class, while a non-static method can be called only from within the class.

Static methods are restricted to accessing static members, whereas non-static methods can access both static and non-static members.

Incorrect



Which coverage type would you use for thorough coverage of FSM states and how would you collect it?

- ☐ Control-oriented functional coverage, asynchronously
- ☒ Data-oriented functional coverage, synchronously
- ☐ FSM coverage, asynchronously
- ☐ FSM coverage, synchronously

Thorough coverage of FSM states is best achieved with FSM coverage collected synchronously.

Incorrect



What is the role of the verification plan in the MDV methodology?

- ☐ To provide stimulus goals, checker requirements, and coverage measures
- ☒ To reduce the area of the design
- ☐ To increase the performance of the design
- ☐ To optimize the design for power consumption

The verification plan in MDV provides stimulus goals, checker requirements, and coverage collection to guide the verification process.

Incorrect



What is the purpose of the following declaration?

```
typedef enum bit[2:0] {ADDI, SUBI, ANDI, XORI, JMP, JMPC, CALL} op_t;
```

- ☒ To declare a set of enumerated types for opcodes
- ☐ To create a list of register identifiers
- ☐ To initialize a set of logic values
- ☐ To define a set of integer constants

This declaration defines a set of enumerated types representing different opcodes.

Correct



Directed tests are the most effective method for finding hidden bugs in modern designs.

- ☐ True
- ☒ False

Directed tests are not the most effective method for finding hidden bugs in modern designs due to their limited ability to cover all possible scenarios.

Correct



What is the purpose of assumptions in formal verification?

- ☐ To create testbenches
- ☒ To exclude illegal input combinations
- ☐ To drive the inputs randomly
- ☐ To prove assertions

Assumptions in formal verification are used to model the environment and exclude illegal input combinations.

Correct



Toggle coverage is the only code coverage available for a gate-level netlist.

- ☐ True
- ☒ False

Toggle coverage is used to verify that the design interconnect is connected and exercised, and it is the only code coverage available for a gate-level netlist.

Incorrect



What is the primary benefit of using 2-state data types in SystemVerilog?

- ☐ Increased power consumption
- ☒ Improved simulation efficiency at higher abstraction levels
- ☐ Reduced performance
- ☐ Larger area

2-state data types in SystemVerilog enhance simulation performance at higher abstraction levels by containing only 0 and 1 values.

Correct

What is the role of a UVM driver?

- ☐ To generate stimulus for the DUT
- ☒ To monitor the DUT
- ☐ To check the outputs of the DUT
- ☐ To apply stimulus to the DUT

The driver in UVM is responsible for applying stimulus to the DUT.

Incorrect

What are the limitations of verification tools as mentioned in the presentation? Select all that apply.

- ☐ They cannot guarantee completeness
- ☒ They can automate all tasks
- ☐ They cannot read minds
- ☐ They can only manage data

Verification tools cannot read minds or guarantee completeness. They can help manage data and automate repetitive tasks, but they cannot ensure that all required functionality is covered due to human error and oversight.

Incorrect



What is the significance of the initial state in formal verification?

- ☐ It influences which states are reachable
- ☒ It is ignored by the verification tool
- ☐ It determines the final state
- ☐ It is always the reset state

The initial state in formal verification influences which states are reachable during the verification process

Incorrect



Which of the following is a concurrent assertion?

- ☐ if (wr_en && rd_en) \$display ("Error");
- ☒ initial forever @(posedge clk) if (wr_en && rd_en) \$display("Error");
- ☐ always @(posedge clk) if (wr_en && rd_en) \$display("Error");
- ☐ assert property (@(negedge clock) !(wr_en && rd_en));

Concurrent assertions are defined with the assert property construct and are evaluated based on a clock event.

Incorrect

Which coverage type measures the execution of code blocks, branches, and expression terms?

- ☐ FSM coverage
- ☒ Code coverage
- ☐ Transaction coverage
- ☐ Functional coverage

Code coverage focuses on the execution of code blocks, branches, and expression terms.

Correct

What is the purpose of the "this" keyword in SystemVerilog?

- ☐ To reference a superclass instance
- ☒ To delete the current class instance
- ☐ To declare a new class instance
- ☐ To reference the current class instance

The "this" keyword is used to reference the current class instance, especially when class properties are hidden by local scope variables.

Incorrect

Which specialized procedural block in SystemVerilog is used for modeling combinational logic?

- ☐ always_latch
- ☒ always
- ☐ always_ff
- ☐ always_comb

The always_comb procedural block in SystemVerilog is used for modeling combinational logic.

Incorrect

What is the main challenge in achieving verification completeness?

- ☐ Complexity of designs
- ☒ Insufficient test coverage
- ☐ Human error and oversight
- ☐ Lack of automation tools

Human error and oversight can lead to incomplete verification, even if all checks pass. This is because the verification plan might not cover all required functionality or might be implemented incorrectly.

Incorrect



What does the following UVM sequence do?

```
class yapp_seq extends uvm_sequence #(yapp_packet);  
  `uvm_object_utils(yapp_seq)  
  function new (string name = "yapp_seq");  
    super.new(name);  
  endfunction  
  virtual task body();  
    `uvm_do(req)  
    `uvm_do_with( req, {addr == 0;} )  
  endtask  
endclass
```

- ☐ Creates two packets, both with random address
- ☐ Creates two packets, both with address 0
- ☐ Creates one random packet with address 0
- ☒ Creates two random packets, the second with address 0

The sequence creates one random packet and then another packet with the address field constrained to 0.

Correct



What is the main difference between formal verification and simulation?

- ☐ Formal verification is based on mathematical proofs
- ☒ Formal verification is intuitive
- ☐ Simulation is exhaustive
- ☐ Formal verification uses testbenches

Formal verification is based on formal logic, using algebraic manipulation of formulas, whereas simulation relies on applying stimulus and observing results.

Incorrect



Code coverage is collected synchronously.

- ☐ True
- ☒ False

Code coverage is collected asynchronously.

Correct



Which constraints can be applied using the with clause in SystemVerilog? Select all that apply.

- ☒ List constraints
- ☐ Relational constraints
- ☐ Conditional constraints
- ☐ Distribution constraints

The with clause can attach various types of constraints, including relational, list, distribution, and conditional constraints.

Incorrect

Which component in UVM is responsible for randomizing sequences and data items?

- ☐ Sequencer
- ☒ Monitor
- ☐ Driver
- ☐ Scoreboard

The sequencer is responsible for randomizing sequences and items in UVM.

Incorrect

Which verification methodology uses constrained random stimulus and coverage measurement?

- ☐ Transaction-based verification
- ☒ Directed test verification
- ☐ Manual verification
- ☐ Metric-Driven Verification (MDV)

Metric-Driven Verification (MDV) uses constrained random stimulus and coverage measurement to improve the predictability, productivity, and quality of the verification effort.

Incorrect



Which keyword is used to declare a static property in a SystemVerilog class?

- ☐ final
- ☒ const
- ☐ static
- ☐ protected

The static keyword is used to declare properties that are shared among all instances of a class.

Incorrect



What does the `assert` directive do in formal verification?

- ☐ It ignores the property
- ☒ It specifies a constraint
- ☐ It demonstrates that an expression can be true
- ☐ It checks that a property is always true

The 'assert' directive in formal verification checks that a property is always true.

Incorrect

Which of the following are key aspects of the verification plan in MDV? Select all that apply.

- ☐ Specifying checks for features
- ☒ Ensuring an acceptable level of coverage
- ☐ Automating test generation
- ☐ Defining features of the design

The verification plan in MDV involves defining the features the design is supposed to have, specifying the checks required to test those features, and ensuring an acceptable level of coverage to be confident that all important scenarios are tested.

Incorrect

Which of the following are key components of the Metric-Driven Verification (MDV) methodology? Select all that apply.

- ☐ Planning verification goals
- ☒ Coverage measurement
- ☐ Generation of random stimulus
- ☐ Checking for correct behavior

The key components of MDV include planning verification goals, generating random stimulus, checking for correct behavior, and measuring coverage.

Incorrect



What is the main use of ICE mode in emulation?

- ☐ To accelerate testbench-driven verification
- ☒ To run non-synthesizable testbenches
- ☐ To verify a design that works with real targets at full speed
- ☐ To compile the design incrementally

ICE mode is used to verify a design that works with real targets at full speed.

Incorrect



SystemVerilog supports multiple inheritance.

- ☐ True
- ☒ False

SystemVerilog supports only single inheritance, meaning each subclass is derived from a single base class.

Correct



Which of the following are verification directives in SystemVerilog Assertions (SVA)? Select all that apply.

- ☐ cover
- ☒ assert
- ☐ restrict
- ☐ assume

The verification directives in SVA include assert, assume, cover, and restrict.

Incorrect



What is a Counter Example (CEX) in formal verification?

- ☐ A proof that a property is always true
- ☒ A waveform showing a failure of an assertion
- ☐ A testbench for simulation
- ☐ A type of assumption

A Counter Example (CEX) is a waveform that demonstrates how an assertion is violated by the design.

Correct

Given the following code, what is the initial value of state?

```
typedef enum logic[2:0]
{
  idle   = 3'b001,
  start  = 3'b010,
  pause  = 3'b100
} onehot_t;

onehot_t state;
```

- ☐ 3'b0000
- ☒ 3'b001
- ☐ 3'b100
- ☐ 3'bxxx
- ☐ 3'b010

The enum onehot_t has an explicit base type logic [2:0]. logic is a 4-state type and so initializes to 3'bxxx.

Incorrect

What is the primary purpose of SystemVerilog?

- ☐ To reduce the power consumption of the design
- ☒ To minimize the area of the design
- ☐ To increase the performance of the design
- ☐ To define a unified language for both design and verification

SystemVerilog aims to define a concise, unified language for design and verification using a single simulation tool.

Incorrect



What is the default base type of a SystemVerilog `enum`?

- ☐ A logic vector sized to the number of enumerated values
- ☒ integer
- ☐ An enum is a unique type with no base type
- ☐ A bit vector sized to the number of enumerated values
- ☐ int

The default type of an enum is always an int - a 32-bit 2-state type. This is why for RTL code, you should always specify an explicit 4-state base type.

Incorrect



Which of the following is an example of a conditional property using same cycle implication?

- ☐ `@(negedge clk) (req && !ack) ##1 bsy;`
- ☒ `@(negedge clk) (req && !ack) |-> bsy;`
- ☐ `@(negedge clk) (req && !ack) |=> bsy;`
- ☐ `@(negedge clk) (req && !ack) && bsy;`

The `|->` operator is used for same cycle implication in conditional properties.

Correct

Which Cadence® tool is used for formal verification?

- ☐ Protium™
- ☒ Xcelium™
- ☐ Virtuoso®
- ☐ Jasper®

Jasper is Cadence's tool for formal verification.

Incorrect

Which of the following statements about inheritance in SystemVerilog is true?

- ☐ A subclass can inherit multiple base classes.
- ☒ A subclass can inherit base classes only if they are declared as final.
- ☐ A subclass cannot inherit any base class.
- ☐ A subclass can inherit only one base class.

SystemVerilog supports single inheritance, meaning each subclass can inherit from only one base class.

Incorrect



What is UVM?

- ☐ A methodology for using verification building blocks
- ☒ An IEEE, non-proprietary standard
- ☐ A class library of verification building blocks
- ☐ All of the above

UVM is a library of SystemVerilog classes, a methodology for using these classes, and is defined as an IEEE standard.

Incorrect



What is the difference between a local and a protected class member in SystemVerilog?

- ☒ Local members are visible only inside the class, while protected members are visible inside the class and its subclasses.
- ☐ Local members are visible to all classes, while protected members are visible only inside the class.
- ☐ Local members are visible inside the class and its subclasses, while protected members are visible only inside the class.
- ☐ Local members are visible only inside the class, while protected members are visible to all classes.

Local members are restricted to the class itself, whereas protected members can be accessed by subclasses.

Correct



What is the role of the `property` keyword in SystemVerilog assertions?

- ☐ To generate a clock signal
- ☒ To declare a new module
- ☐ To define a design behavior to be checked
- ☐ To declare a variable

The `property` keyword is used to define a design behavior that will be checked by the assertion.

Incorrect

